

NAME: _____ CLASS: _____ INDEX: _____



CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 1

BIOLOGY

Paper 1 Multiple Choice

8875/01

MONDAY 19 SEPTEMBER 2011

1 Hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write and/or shade your name, NRIC / FIN number and HT group on the Answer Sheet in the spaces provided unless this has been done for you.

There are **30 MCQ** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

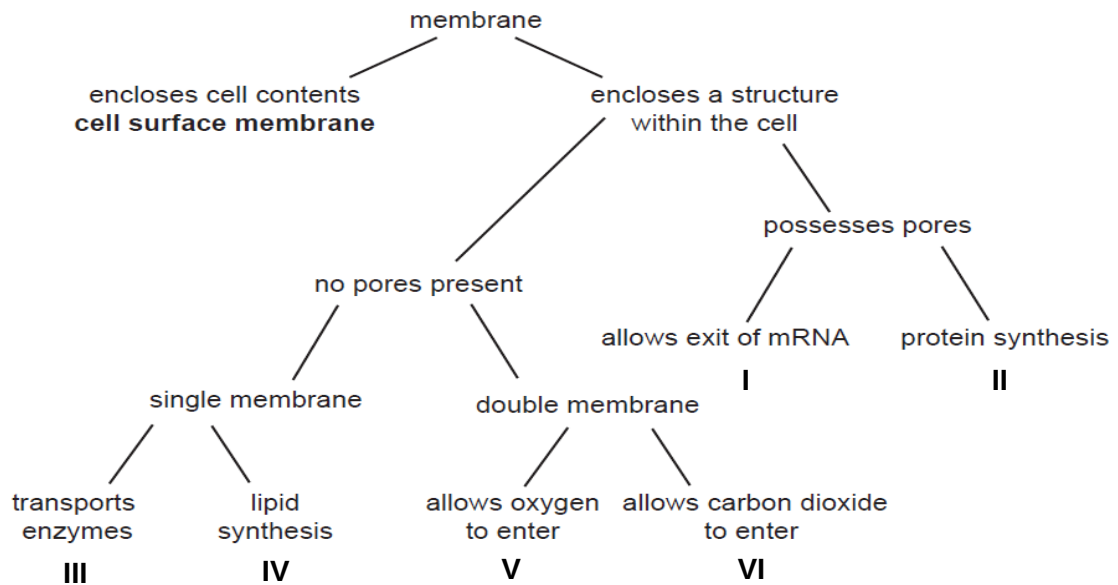
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

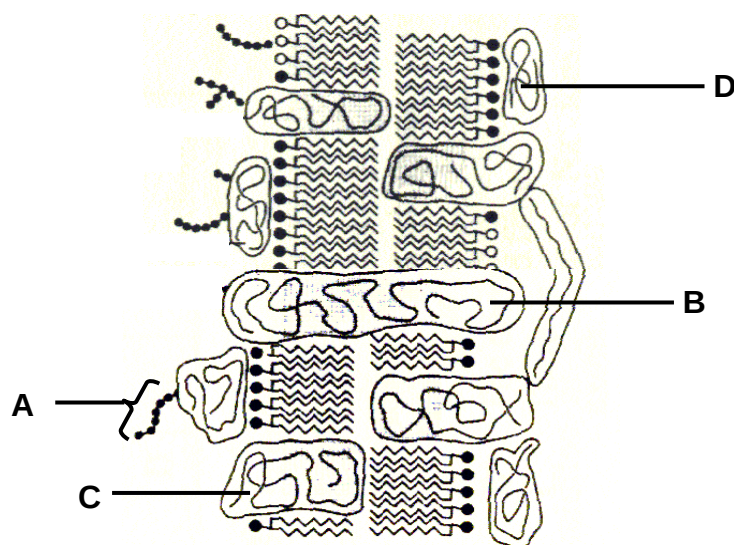
1. Membranes within and at the surface of cells have different roles. The flowchart shows a scheme to identify various organelles within the cell, by describing the structure and function of their membranes.



Which of the following outcomes correctly identifies the various cellular organelles?

	I	II	III	IV	V	VI
A	chloroplast	vesicle	smooth ER	rough ER	nucleolus	mitochondrion
B	nucleolus	rough ER	vesicle	smooth ER	nucleus	mitochondrion
C	nucleus	smooth ER	mitochondrion	rough ER	vesicle	chloroplast
D	nucleus	rough ER	vesicle	smooth ER	mitochondrion	chloroplast

2. The diagram below shows a section of a cell surface membrane.

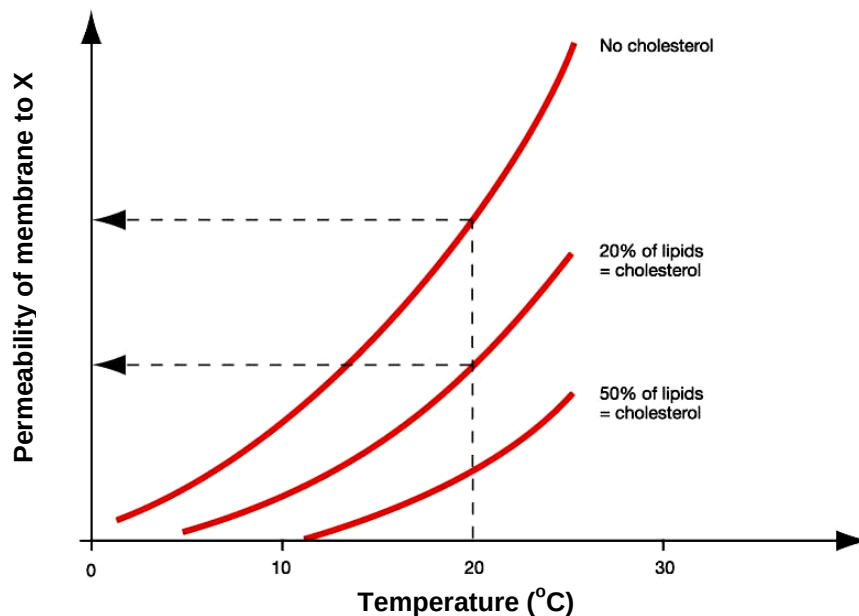


Which of the following statements are correct?

- I Structure **A** is found only on the extracellular surface of the membrane.
- II Structure **B** may contain a channel that is hydrophobic to allow ions to move across the membrane.
- III Structure **B** is able to bind to certain substances and change its conformation to transport these substances into the cell.
- IV Structure **C** is able to change the structure of substances such that the substance can now diffuse across the membrane
- V Structure **D** is found only on one surface of the membrane.

- A I, II, III and IV
- B II, III and V
- C I and III only
- D I and V only

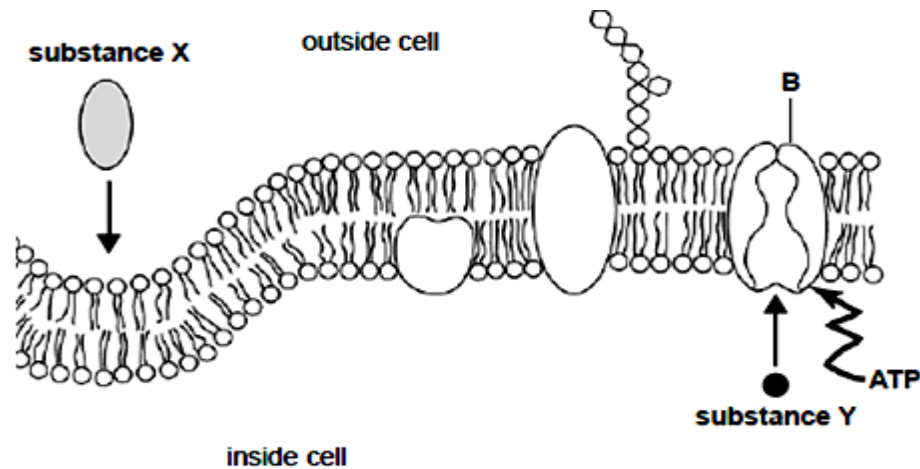
3. The graph below shows the permeability of three different membranes to **chemical X** at different temperatures. These three membranes differ in the amount of cholesterol present in the phospholipid bilayers.



Which of the following is a possible explanation for the observed data?

- A Increase in temperature increases the permeability of the membrane to **chemical X** as cholesterol increases the fluidity of the phospholipids.
- B At 20°C, an increase in the proportion of cholesterol in the membrane increases the permeability of membrane to **C** as both cholesterol and **chemical X** are non-polar.
- C Increase in the proportion of cholesterol decreases the permeability of the membrane to **G** as cholesterol decreases the fluidity of the phospholipids.
- D With an increase in the proportion of cholesterol in the membrane, a lower temperature is required to achieve the same level of permeability for **chemical X** as **chemical X** will gain a higher kinetic energy to penetrate the membrane.

4. The figure below shows a cross section of the plasma membrane in a typical mammalian cell. The substances labelled **X** and **Y** are about to be transported across the membrane in the directions shown by the arrows (→).



Which of the following is correct?

	X			Y		
	<i>type of movement</i>	<i>nature of molecule</i>	<i>size of molecule</i>	<i>type of movement</i>	<i>nature of molecule</i>	<i>size of molecule</i>
A	diffusion	lipid soluble	small	facilitated diffusion	lipid soluble	large
B	diffusion	water soluble	small	active transport	lipid soluble	small
C	endocytosis	lipid soluble	large	facilitated diffusion	charged	large
D	endocytosis	water soluble	large	active transport	charged	small

5. Most wild plants contain toxins that deter animals from eating them. A scientist discovered that a toxin produced by a certain plant was also toxic to the same plant if it was applied to the roots of the plant. As the first step on finding out why the plant was not normally killed by its own toxin, he fractionated some plant cells and found that the toxin was in the fraction that contained the largest cell organelle. He also found that the toxin was no longer toxic after it was heated.

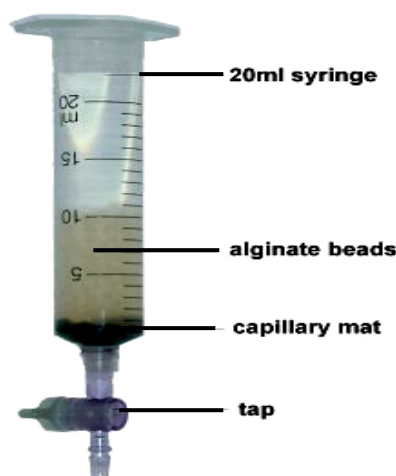
Which of the following statements are consistent with the scientist's observations?

- I** The toxin was stored in the central vacuole.
 - II** The toxin cannot cross the membrane of the organelle in which it is stored.
 - III** The toxin was stored in chloroplast.
 - IV** The toxin is likely to be lipid-soluble.
 - V** The toxin may be an enzyme.
- A** I, II and V
 - B** I, IV and V
 - C** II, III and IV
 - D** III, IV and V

6. Which of the following best accounts for the difference in structure between amylose which is helical and cellulose which is fibrous?
- A Presence or absence of 180° rotation of alternate subunits.
 - B Presence or absence of branching in the macromolecule
 - C Subunits are either monosaccharides or amino acids.
 - D Different tendency of macromolecule to form hydrogen bonds with water.
7. The **R** group of the amino acid *serine* is $-\text{CH}_2-\text{OH}$. The **R** group of the amino acid *alanine* is $-\text{CH}_3$.
Where would you expect to find these amino acids in a globular protein in aqueous solution?
- A Serine would be in the interior, and alanine would be on the exterior of the globular protein.
 - B Alanine would be in the interior, and serine would be on the exterior of the globular protein.
 - C Both serine and alanine would be in the interior of the globular protein.
 - D Both serine and alanine would be in the interior and on the exterior of the globular protein.
8. One complete turn of the double helix of DNA contains 10 pairs of bases and is 3.4 nm long. What is the approximate length of the DNA coding sequence of lysozyme, a protein of 129 amino acids?
- A 66 nm
 - B 132 nm
 - C 70 nm
 - D 140 nm
9. Which of the following statement is true for enzyme inhibitors?
- 1 Inhibitors that form covalent bonds with enzymes often inactivate the enzymes permanently.
 - 2 They often target enzymes at the beginning of a metabolic pathway in end product inhibition.
 - 3 Enzyme inhibitors repress the transcription of the genes coding for enzymes.
 - 4 Post translational modifications are an effective way to regulate the activity of the inhibitors.
- A 2 and 4
 - B 2, 3 and 4
 - C 1, 2 and 4
 - D 1, 2, 3 and 4

10. Molecules of a stomach enzyme were immobilised (strongly attached) within alginate beads and used in an investigation to determine the identity of four different unknown substances (**W**, **X**, **Y**, **Z**) that affected the enzyme activity even when its substrate was present in excess.

The beads were packed into a column set up similar to the one shown below.



In order to allow the enzyme activity to equilibrate, 10ml of substrate solution was poured in from the top and allowed to drain through the alginate bead column thrice before the experiment was started.

Following the equilibration, 10ml of unknown substance **W** was allowed to drain through the alginate bead column fully *before* another 10ml of substrate solution was drained through the column to allow the enzymes to react with the substrates. The concentration of product obtained in the filtrate was measured to reflect the level of enzyme activity after addition of the unknown substance. These steps were then repeated with the other three unknown substances added into the same alginate bead column in the order of **X**, **Y**, then **Z**.

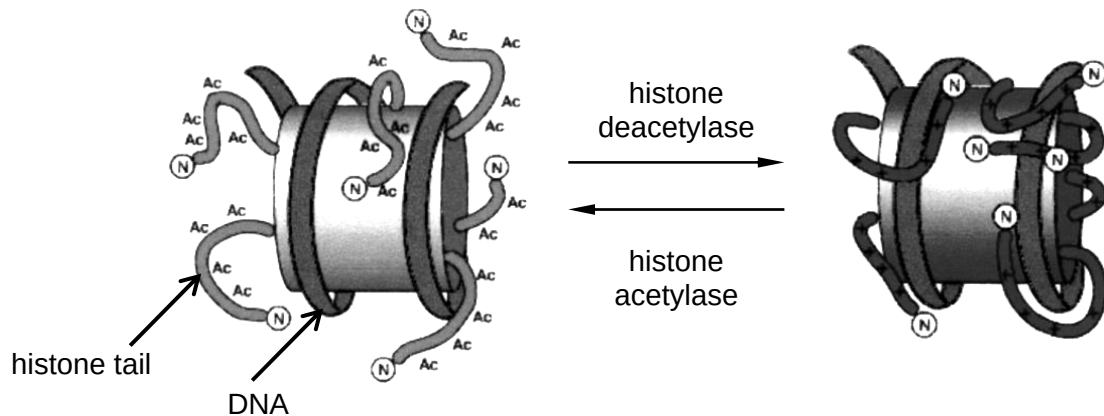
The table below shows the results obtained.

Unknown substance	Enzyme activity after adding unknown substance (arbitrary units)
W	20.7
X	8.5
Y	0.1
Z	0.1

Given that the enzyme activity in the absence of any unknown substance was 16.8 arbitrary units after equilibration, which combination correctly identifies the four unknown substances based on the results obtained?

	W	X	Y	Z
A	cofactor	pH buffer at pH 2	competitive inhibitor	non-competitive inhibitor
B	substrate	cofactor	pH buffer at pH 7	competitive inhibitor
C	cofactor	non-competitive inhibitor	pH buffer at pH 13	substrate
D	substrate	competitive inhibitor	non-competitive inhibitor	cofactor

11. The diagram below shows how acetylation of histones promotes loose chromatin structure. Recent evidence has shown that chemical modification of histones play a direct role in regulation of gene expression.



Which of the following best explains how acetylation regulates gene expression?

- A Helicase action is enhanced by acetylation.
 - B Acetylation of histones neutralizes their negative charges and encourages binding to DNA polymerase.
 - C When nucleosomes are highly acetylated, chromatin becomes less compact and DNA is more accessible for transcription.
 - D RNA polymerase works better by binding with acetyl groups.
12. Which of the following statements are true of the end-replication problem of DNA?
- I When a linear DNA molecule replicates, a gap is left at the 3' end of each new strand because DNA polymerase can only add nucleotides to a 5' end.
 - II Telomerase prevents the end-replication problem from occurring.
 - III Repeated rounds of replication produce shorter and shorter DNA molecules.
 - IV Prokaryotes do not have the end-replication problem.
- A I and II
 - B I and IV
 - C II and III
 - D III and IV

13. The following events occur in the extension of telomeres.

- 1 Further extension of 3' end of telomere.
- 2 Translocation of telomerase to end of telomere.
- 3 Reverse transcription to extend the 3' end of the telomere.
- 4 Complementary base-pairing of RNA template with single-stranded end of telomere.

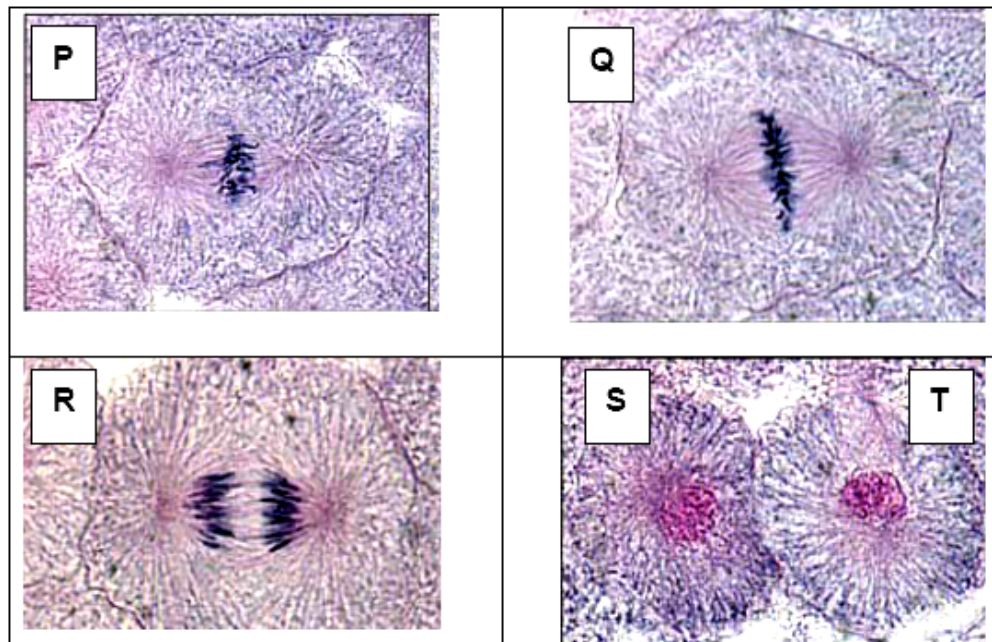
In which order do these events take place?

- A 3 2 4 1
- B 3 4 2 1
- C 4 2 3 1
- D 4 3 2 1

14. Which of the following shows the possible effects of a single nucleotide substitution in each of the following locations in a gene on the production of the protein it codes for?

	promoter	transcription terminator	start codon	stop codon	middle of an intron
A	No protein product is produced	Protein product is shorter than normal	Protein product is longer than normal	Protein product is normal	Too much protein product is produced
B	Too much protein product is produced	Protein product is normal	No protein product is produced	Protein product is longer than normal	Protein product is normal
C	Protein product is normal	Protein product is longer than normal	Protein product is shorter than normal	Too much protein product is produced	Protein product is longer than normal
D	Protein product is longer than normal	Too much protein product is produced	Protein product is normal	Protein product is shorter than normal	No protein product is produced

15. Cells **P**, **Q** and **R** are undergoing various stages of mitosis. Cells **S** and **T** are daughter cells formed immediately after cell division.



The plant tissue had been growing in a culture medium containing radioactive DNA nucleotides. The level of radioactivity in each cell was measured and the presence of sister chromatids were noted. Cells with the same amount of DNA have approximately the same level of radioactivity.

Which cell has a level of radioactivity and observations that could have been resulted due to inaccurate readings and/or observations?

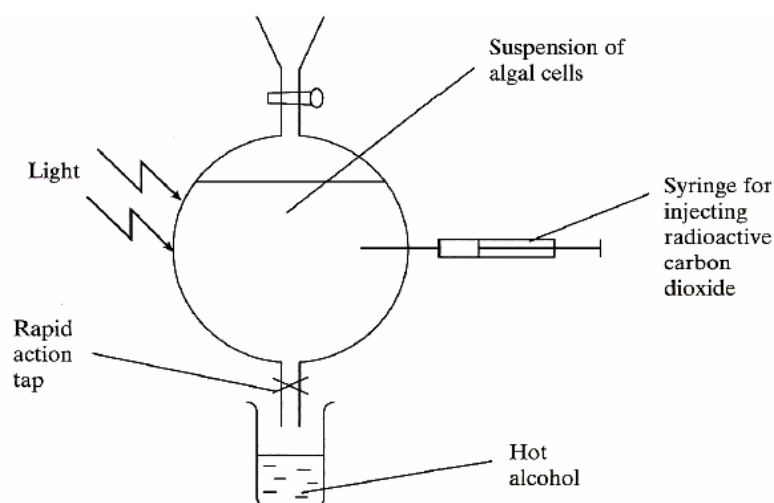
	Cell	Radioactivity level (a.u.)	Presence of sister chromatids
A	P	102	Yes
B	Q	101	Yes
C	R	52	No
D	S	51	No

16. Which statement about cancer is not correct?

- A Both alleles for a tumour suppressor gene must undergo loss-of-function mutations in the development of cancer.
- B Mutations in proto-oncogenes may not necessarily result in excessive cell proliferation.
- C The growth of new blood vessels is a fundamental step in the development of a malignant tumour.
- D The risk of cancer is higher when *p53* gene is placed under the control of a strong viral promoter.

17. An investigation was carried out to find out the sequence of biochemical changes that occur during photosynthesis.

Radioactive carbon dioxide was added to a suspension of algal cells, and they were allowed to photosynthesise. At intervals, samples of the suspension were removed into hot alcohol. These samples were analysed for different radioactively labelled compounds. The experimental set up is shown below.



Samples were removed from the suspension at five different times, between 5 seconds and 600 seconds after the start of the experiment. In each sample, the amount of radioactivity present in four different organic compounds, **P**, **Q**, **R** and **S** was measured. The table shows the results.

organic compound	amount of radioactivity present / arbitrary units				
	5 s	15 s	60 s	180 s	600 s
P	0.01	0.02	0.08	0.17	0.67
Q	1.00	2.00	3.10	3.15	3.15
R	0.10	1.50	2.20	2.30	2.40
S	0.05	0.11	0.16	1.00	1.00

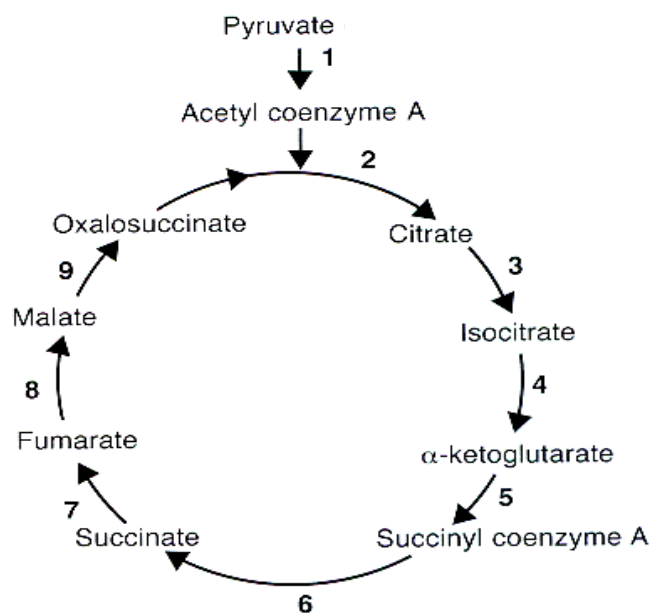
What are the most likely identities of the four organic compounds?

	P	Q	R	S
A	TP (GALP)	GP	glycerate 1,3-bisphosphate	RuBP
B	GP	RuBP	TP (GALP)	glycerate 1,3-bisphosphate
C	RuBP	GP	glycerate 1,3-bisphosphate	TP (GALP)
D	TP (GALP)	RuBP	GP	glycerate 1,3-bisphosphate

18. Dinitrophenol is a metabolic poison that can lodge within the thylakoid membranes of chloroplasts. It then provides an alternate route for H^+ ions to diffuse across the thylakoid membranes. In what way will the Calvin cycle (dark reaction) be affected in chloroplasts poisoned with dinitrophenol?

- A** No effect since Calvin cycle is an enzyme-controlled process.
- B** The rate of Calvin cycle will increase because pH in the stroma will drop towards the optimum.
- C** The rate of Calvin cycle will decrease with the accumulation of glycerate-3-phosphate (GP).
- D** The rate of Calvin cycle will decrease with the accumulation of triose phosphate (TP).

19. At which stages are hydrogen atoms transferred to NAD^+ ?



- A** 1, 4, 5, and 7
- B** 1, 4, 5, and 9
- C** 2, 4, 5, and 7
- D** 4, 5, 7, and 9

20. Two test tubes containing the following contents are shown below:

Tube 1: Radioactive glucose solution + yeast cells suspension + oxygen + antimycin

Tube 2: Radioactive glucose solution + yeast cells suspension + oxygen

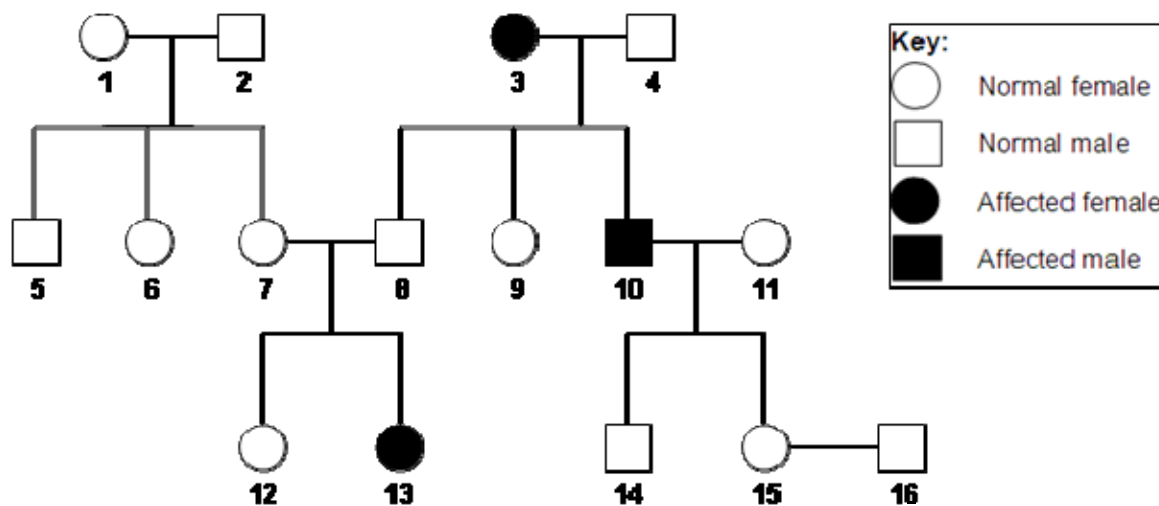
Radioactive glucose has all its six carbons made of radioactive ^{14}C . The initial radioactivity measured for the glucose in each test tube is 60 arbitrary units.

Antimycin is an electron transport chain inhibitor.

If the gaseous product and the aqueous products are tested using a radioactive meter after all the glucose has been metabolized, what would be the final observed readings?

	Tube 1 (radioactivity measured / a.u.)		Tube 2 (radioactivity measured / a.u.)	
	aqueous products	gaseous products	aqueous products	gaseous products
A	0	60	40	20
B	20	40	0	60
C	40	20	0	60
D	40	20	60	0

21. The pedigree shows the inheritance of a condition known as galactosemia.



What is the probability that the first child of individuals 15 and 16 will be a normal boy?

- A 3/4
- B 3/8
- C 7/8
- D 7/16

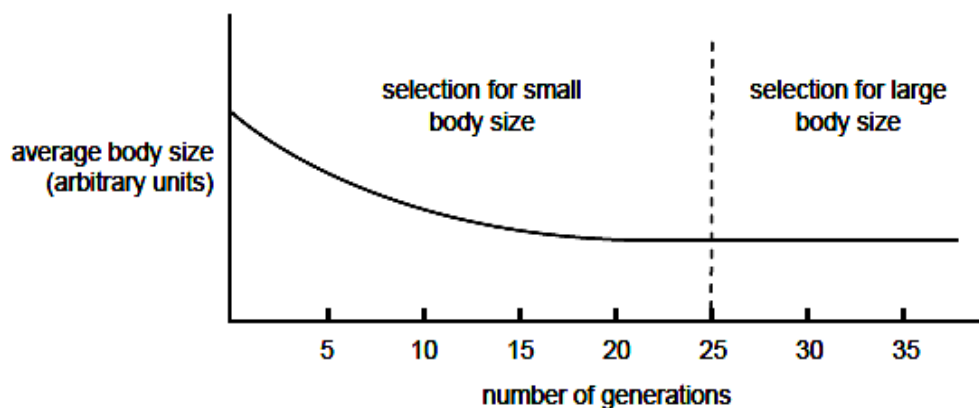
22. In pea plants, the allele (Y) coding yellow seeds is dominant to the allele (y) coding green seeds. The allele (R) coding for round seeds is dominant to the allele (r) coding for wrinkled seeds. Plants grown from yellow, round seeds were crossed with plants grown from green, wrinkled seeds.

The table below shows the phenotype of 80 offspring plants.

Number of offspring	Seed colour	Seed shape
23	yellow	round
20	yellow	wrinkled
17	green	round
20	green	wrinkled

Which of the following sets of parental genotypes would produce the observed results above?

- A $YYRR \times yyrr$
 B $YyRR \times yyrr$
 C $YyRr \times yyrr$
 D $YyRr \times YyRr$
23. The graph shows the results of a breeding experiment with the vinegar fly. In each of the first 25 generations the smallest flies were selected to produce the next generation. After 25 generations the selection was reversed. From generations 25 to 35, the largest flies were chosen to breed the next generation.



With respect to the genes controlling body size, the results of the experiment suggest that

- A after 25 generations, there was no genetic variation.
 B for the first 25 generations, there was no genetic variation.
 C selection between generations 25 and 35 had a significant effect on average body size.
 D if selection for small body size continued after generation 25, average body size would continue to fall.

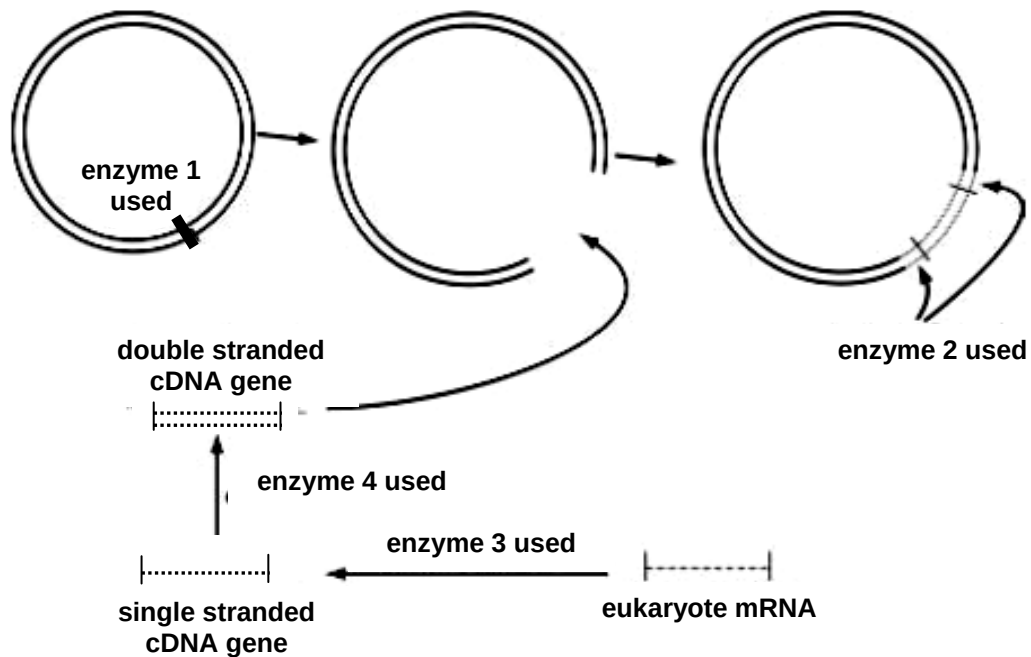
24. Some statements concerning evolution are listed:

- 1 Offspring tend to resemble their parents.
- 2 Individuals in a sexually reproducing population are different.
- 3 The fossil record shows that many species have become extinct.
- 4 More offspring are produced than can possibly survive to sexual maturity.
- 5 Characteristics acquired during an organism's lifetime are passed to its offspring.

Which of these statements form the basis of Darwin's theory of evolution by natural selection?

- A 1, 2 and 3
 B 1, 2 and 4
 C 1, 3 and 5
 D 1, 4 and 5

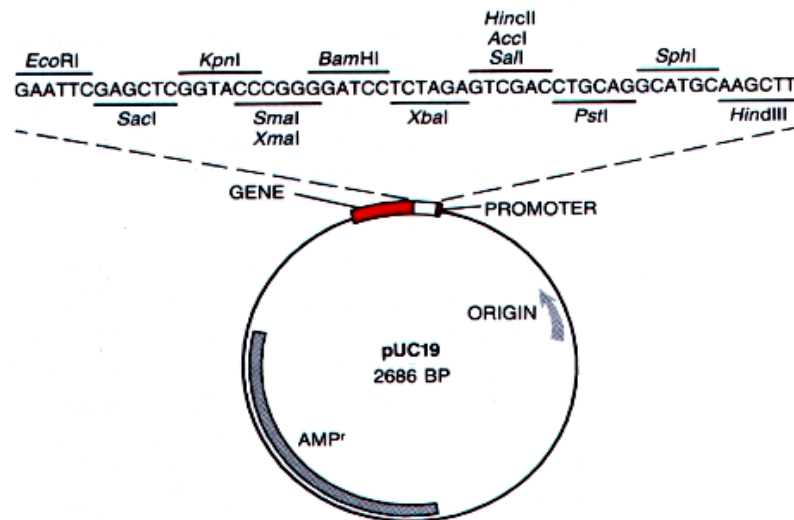
25. The diagram shows part of a process used for genetic engineering.



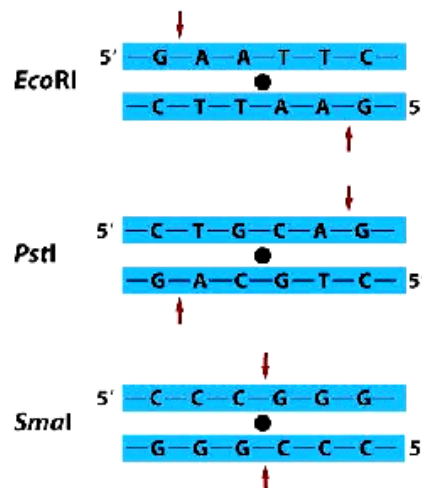
Which of the following shows the correct identity of the enzymes used in this part of the process?

Enzyme				
	1	2	3	4
A	Restriction enzyme	DNA ligase	RNA polymerase	Reverse transcriptase
B	Restriction enzyme	DNA ligase	Reverse transcriptase	DNA polymerase
C	DNA ligase	Restriction enzyme	DNA polymerase	Reverse transcriptase
D	Restriction enzyme	Reverse transcriptase	RNA polymerase	DNA ligase

26. The figure shows the plasmid map of pUC19, which includes the positions of the origin of replication (ORI), multiple cloning sites (MCS) and two genes - *amp^R* and *lacZ*. pUC19 is used for the production of human anti-thrombin III (hATIII) in the bacterium *E. coli*.



The restriction sites of three particular restriction enzymes, *EcoRI*, *PstI* and *SmaI* are shown below.



Which of the following can be concluded from the figures?

I	Any restriction enzyme whose restriction site is found within the MCS can be used to cleave pUC19.
II	pUC19 cleaved with <i>EcoRI</i> can anneal to hATIII cDNA cleaved with <i>PstI</i> as both restriction enzymes generate single-stranded overhangs of the same length.
III	The use of <i>SmaI</i> will require an additional step of ligating adaptors to the ends of pUC19 and hATIII cDNA.
IV	Colonies of <i>E. coli</i> transformed with recombinant pUC19 will survive on agar plates containing ampicillin and X-gal, giving a blue appearance.

- A I and III only
 B I and IV only
 C II and III only
 D All of the above.

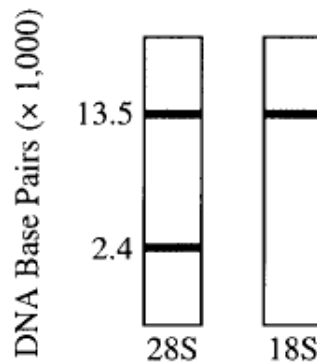
27. How is the polymerase chain reaction (PCR) similar to the replication of DNA?

- I DNA is heated to break hydrogen bonds
- II DNA unzips
- III free nucleotides are used
- IV DNA polymerase are required

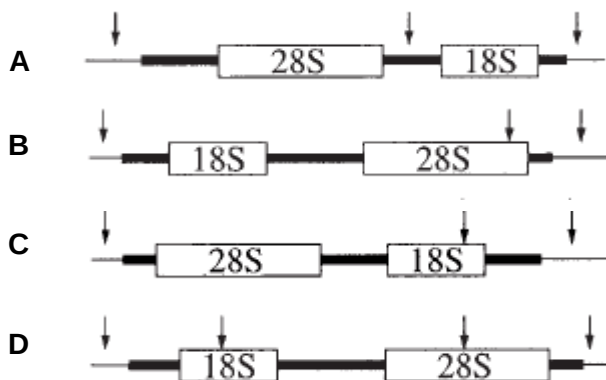
- A I only
- B II and IV only
- C I, II and III only
- D II, III and IV only

28. The autoradiograms obtained below (after electrophoresis and Southern Blotting) show human DNA digested with a specific restriction enzyme and probed with labeled rRNA.

On the left, the probe used was 28S rRNA. On the right, the probe used was 18S rRNA.



If the arrows in the following map show the location of the restriction sites of this restriction enzyme, which map *best* explains the results shown above?



29. In terms of containment (prevention of genetic pollution of closely-related wild plant or non-GM crop populations), which of the following is an advantage of chloroplast transformation (i.e. the introduction of foreign genes into chloroplasts of host plants) over nuclear transformation?
- A Chloroplasts are surrounded by a double membrane.
 - B There are no chloroplasts in pollen of most plant species.
 - C Chloroplasts are smaller than the nucleus.
 - D There is no DNA in chloroplasts.
30. Normal tomato plants have a gene that codes for an enzyme that softens tomatoes as they ripen. The DNA sequence for this gene is as follows:

3' ... T T A G C C T T A ... 5',

Genetically engineered tomatoes ripen and soften more slowly because of the insertion of a transgene which reduces the amount of the softening enzyme produced.

Which of the following would be the mRNA sequence encoded by this transgene?

- A 5' – A U U C C G A U U – 3'
- B 5' – A A T C G G A A T – 3'
- C 5' – A A U C G G A A U – 3'
- D 5' – U U A G C C U U A – 3'

BLANK PAGE